

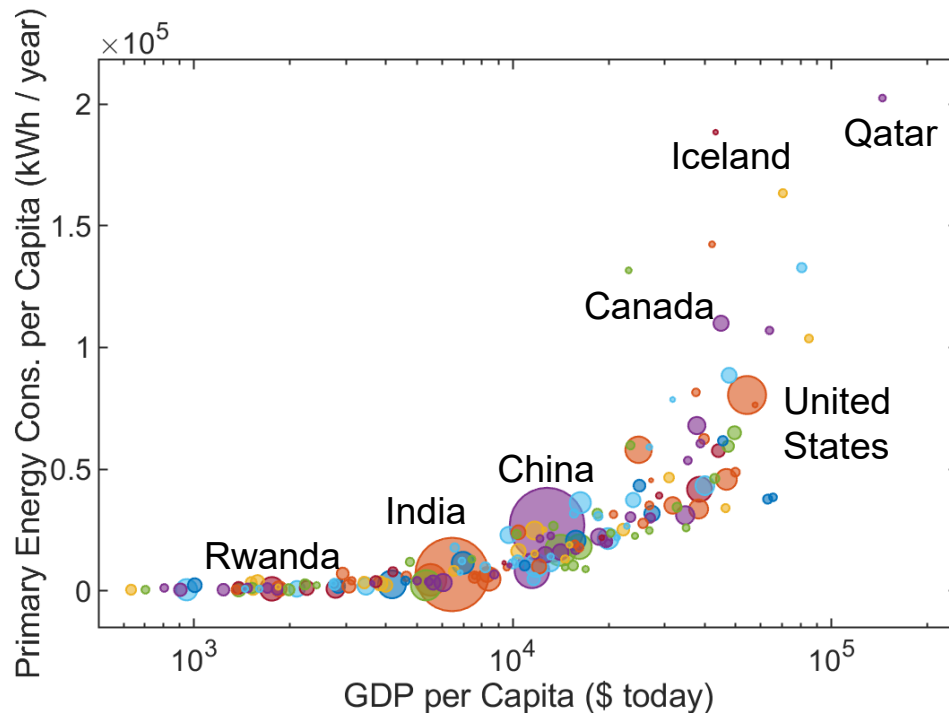
Thermodynamics

Lecture 1B:

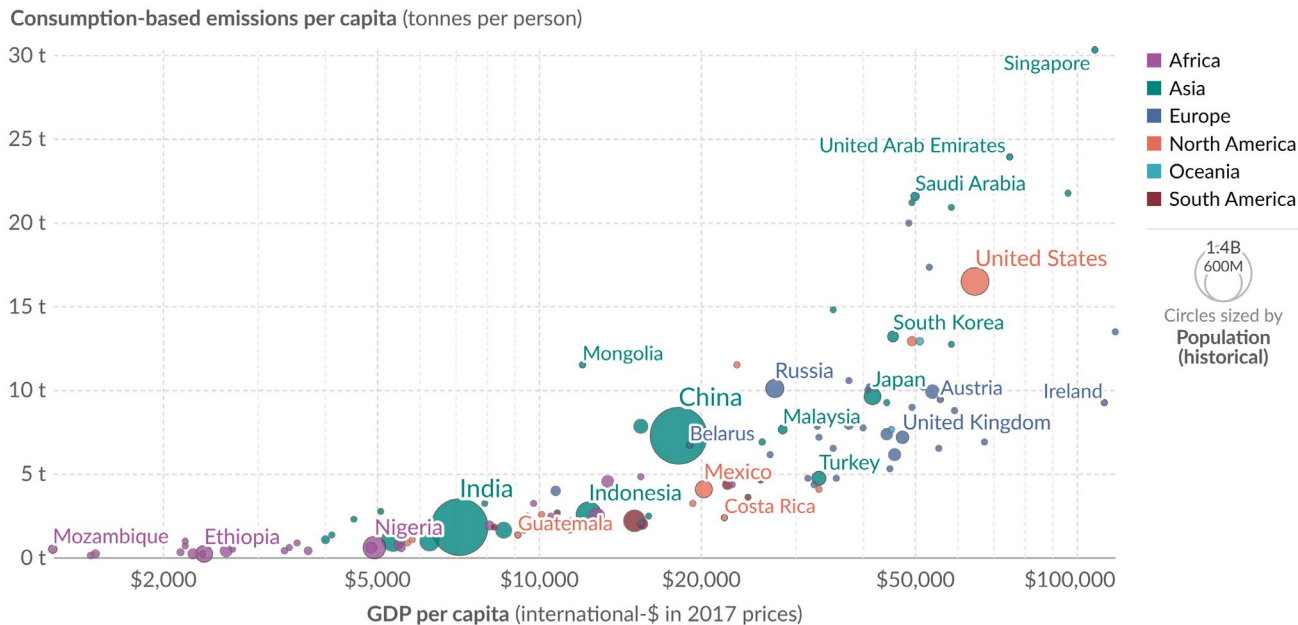
- Motivation
- Basic Concepts
- System properties

Prof. Claudio Hail (he/him)

Global energy demand



Energy and CO₂ emissions



Data source: Global Carbon Budget (2024); Population based on various sources (2024); World Bank (2023)
OurWorldinData.org/co2-and-greenhouse-gas-emissions | CC BY

Thermodynamics: relevance



Airplanes



Power plants



Cars

Thermodynamics: relevance



Food processing



Refrigeration



Heating

Thermodynamics challenges in the future

- Electrification
- Decarbonization
- Artificial Intelligence
- Emerging technologies:
 - Atmospheric water harvesting
 - Passive radiative cooling
 - ...

Learning objectives

What you will be able to do after this course:

- I. *Knowledge:* Write the first and second laws of thermodynamics.**
- II. Comprehension:* Describe the first and second laws in the student's own language.
- III. Application:* Solve simple single-answer problems using the first law.
- IV. Analysis:* Solve problems requiring both the first and second laws.
- V. Synthesis:* Develop schemes to improve the efficiency or performance of thermodynamic systems, e.g., compressors, turbines, heat pumps, etc.

SI Units

Dimension	Unit
Length	meter (m)
Mass	kilogram (kg)
Time	seconds (s)
Temperature	Kelvin (K)
Electric current	Ampere (A)
Amount of light	candela (cd)
Amount of matter	mole (mol)

Base SI units

Derived SI units

Force	Newton (N)
Energy (Work, Heat ..)	Joule (Nm)
Pressure	Pascale (Pa)
Temperature	Celsius (°C)

Any equation: ensure homogeneity of terms

$$P_{tot} = \frac{10 \text{ N}}{2 \text{ m}^2} + \frac{30 \text{ lbf}}{3 \text{ ft}^2}$$

Phases of matter

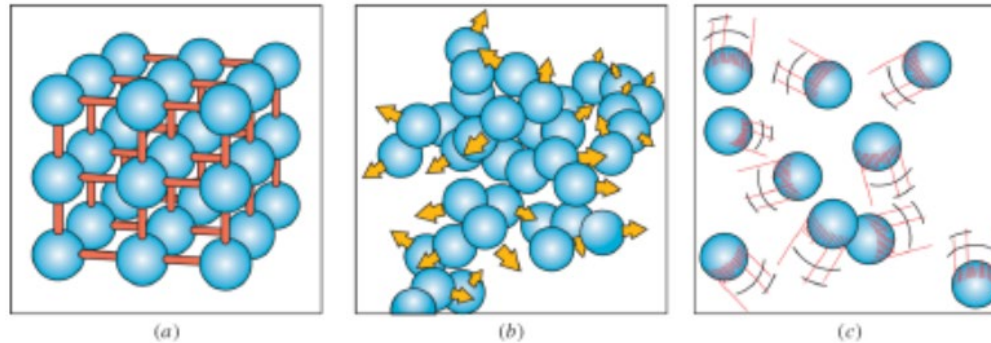


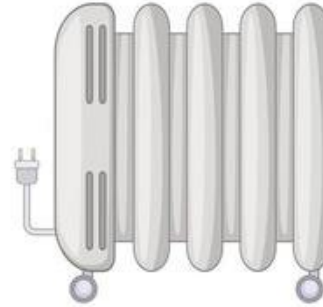
FIGURE 3–4

Thermodynamic systems

Definition: A region of space containing matter in the form of molecules at a given point in time for which a thermodynamic analysis is performed.

- Boundary: physical or imaginary
- Systems and boundaries can change overtime (Moving boundaries)

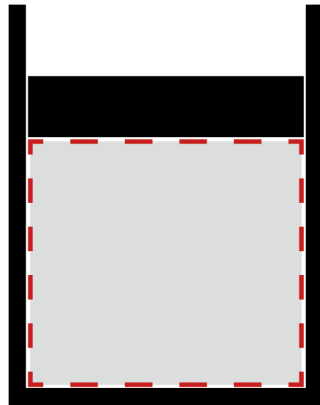
Thermodynamic systems: examples



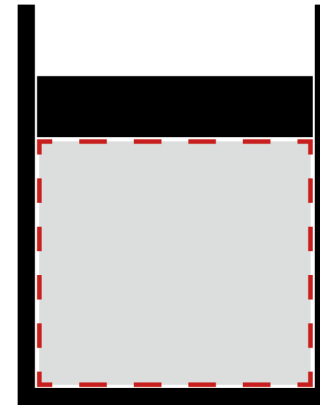
Before any thermodynamic analysis: define system boundary, choosing right system, can simplify analysis!

Thermodynamic systems

- **Closed system:**

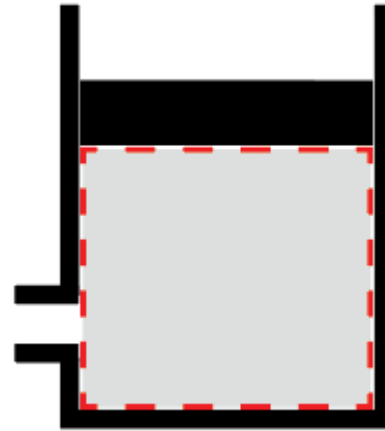
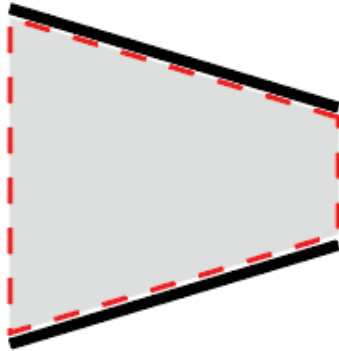


- **Isolated system:**



Thermodynamic systems

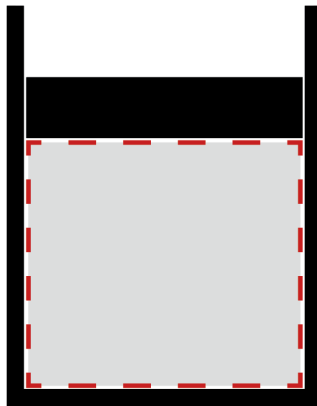
- Open system / Control volume



- Can also change over time, and show interactions W , Q etc.

Thermodynamic state and state properties

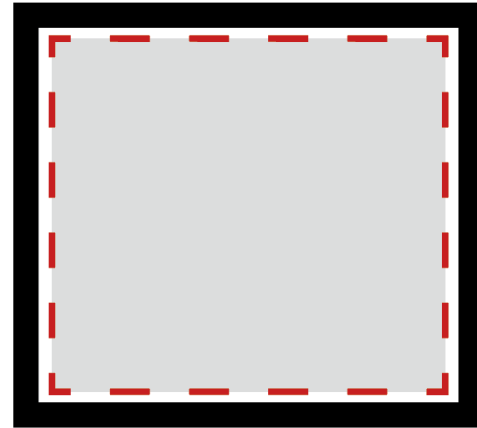
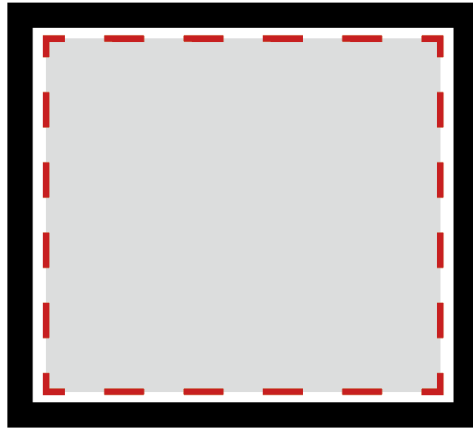
- Definition: state of the system that is not undergoing any change, which is fully defined by thermodynamic properties.



State properties

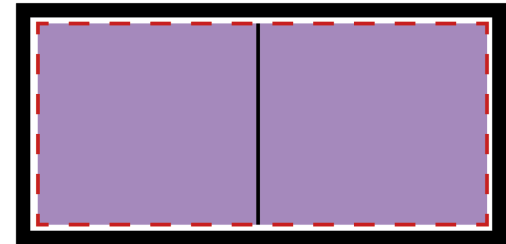
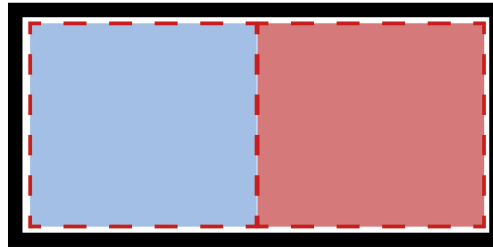
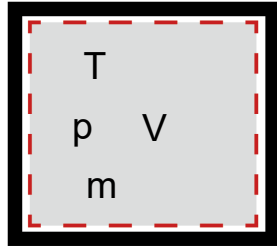
Thermodynamic state properties

- Intensive vs extensive and specific properties



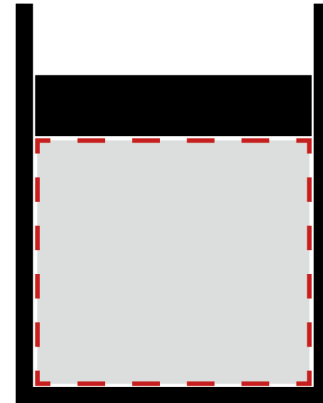
Thermodynamic equilibrium

- Definition: state of balance, no unbalanced potential or driving force in the system.
- System experiences no changes when isolated from surroundings.



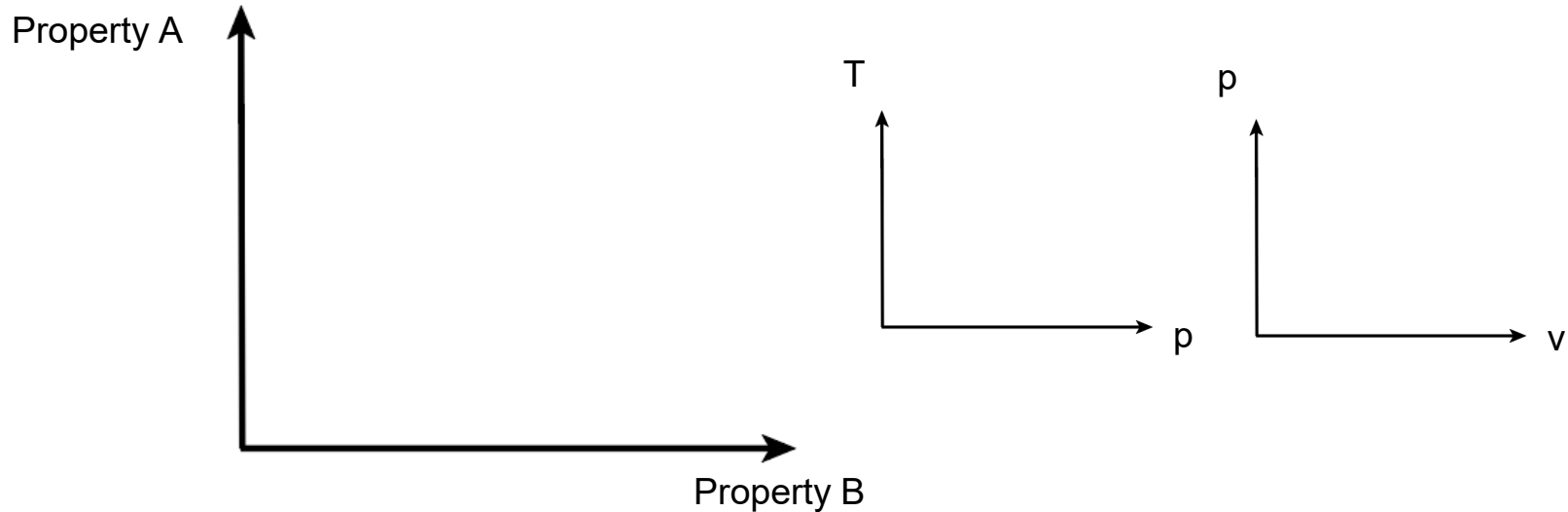
Thermodynamic state postulate

- Definition: A simple compressible system is completely defined by two independent intensive state properties.



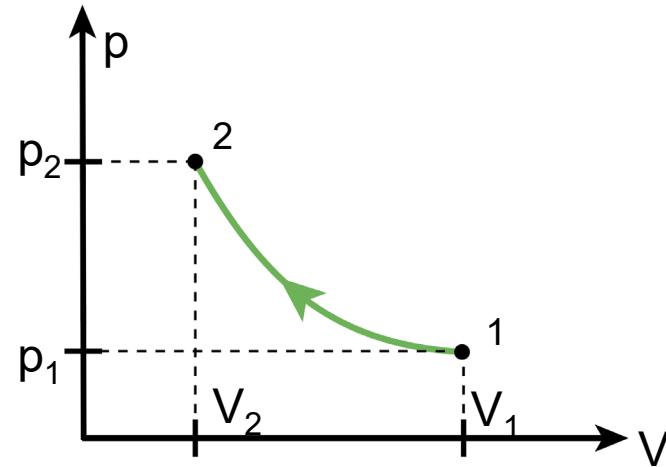
Thermodynamic process (I)

- Definition: Transformation from one state to another through interaction with the surroundings.

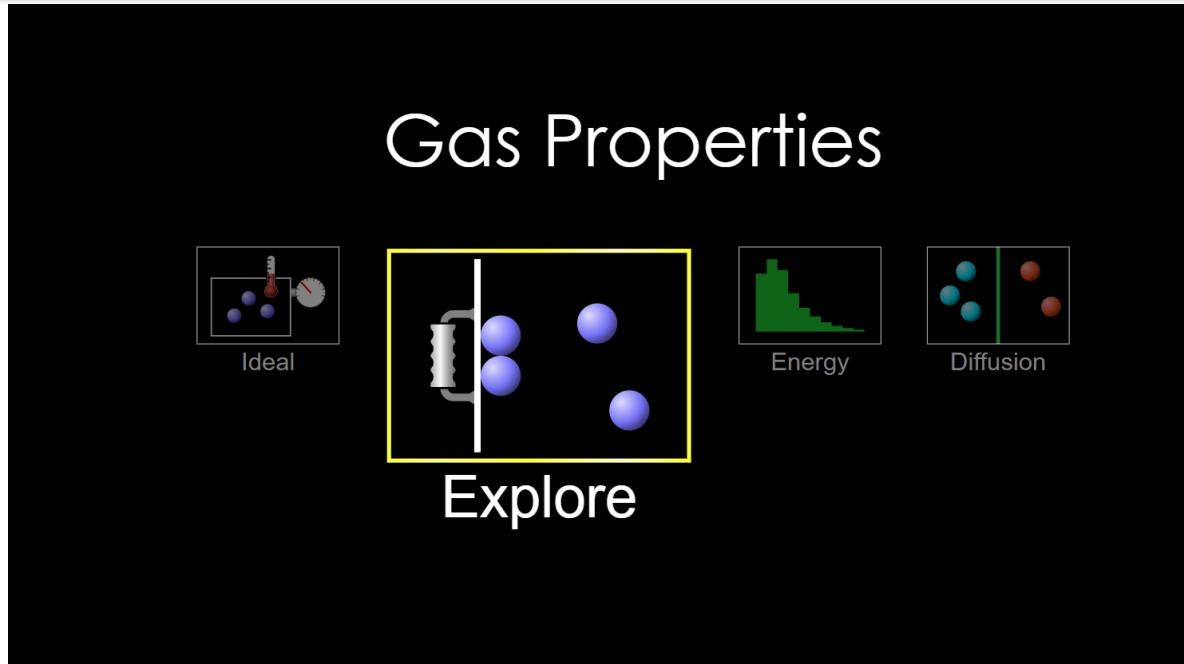


Thermodynamic process (II)

- Quasistatic process:
- Non-quasistatic process:

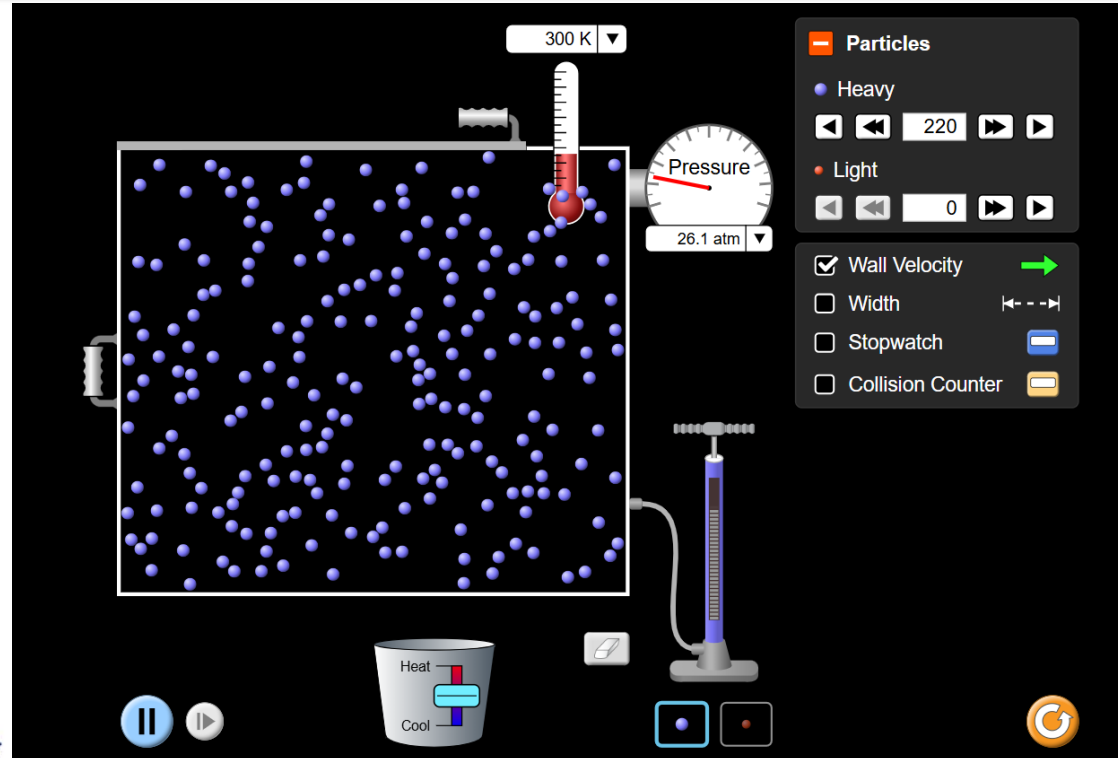


Simulation: Gas in a cylinder



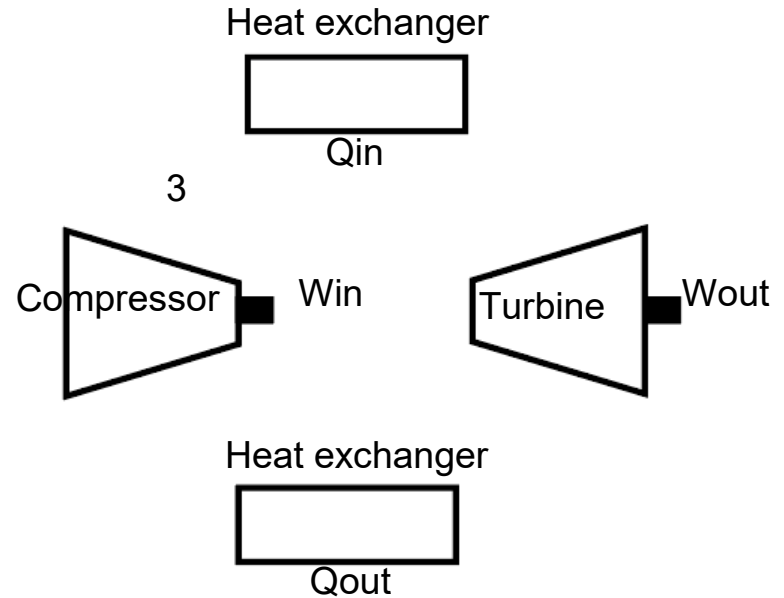
https://phet.colorado.edu/sims/html/gas-properties/latest/gas-properties_all.html

Simulation: Gas in a cylinder



Thermodynamic Cycles

- System undergoing a series of Th. processes and returning to the initial state.



Density

- Density: Mass per volume
- Specific Volume:
- Generally:
- Incompressible fluid:

Temperature

- A measure of how hot or cold
- **Celsius:** ice point 0°C , steam point 100°C
- **Fahrenheit:** ice point w salt 0°F , human body 96°F
- **Kelvin:** absolute zero 0 K

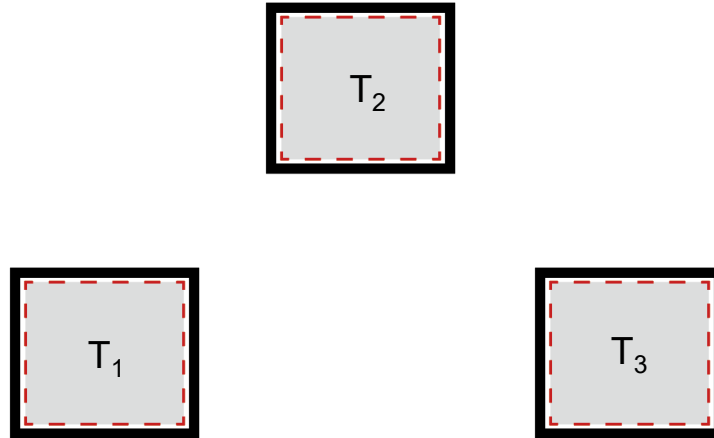
$$T(K) = T(^{\circ}\text{C}) + 273.15$$

$$T(K) = \frac{T(^{\circ}\text{F}) - 32}{1.8} + 273.15$$

$$\Delta T(K) = \Delta T(^{\circ}\text{C})$$

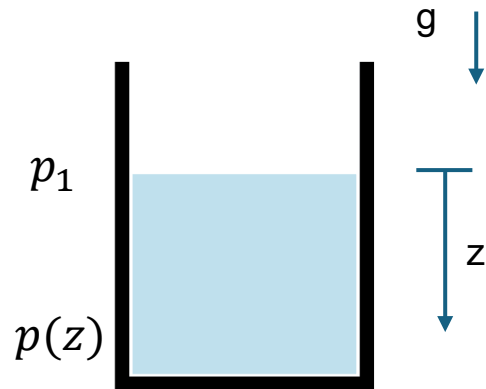
0th law of thermodynamics

- Thermal equilibrium: two objects at the same temperature.
- 0th law:



Pressure

- Normal force per unit area
- Absolute pressure p_{abs} relative to vacuum
- Gauge pressure:
- Vacuum pressure:
- External forces:



Summary

- Learned about thermodynamic systems, states, properties, equilibrium, processes, cycles.
- Density, pressure, temperature

Next week

- Different kinds of energy
- Learn about integral energy
- Energy transfer by work